

The Real Number System

Answer each question. Show your work where directed.

SCORE 3.0 Basic Understanding

1.1 Converting Rational Numbers

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Convert each number as directed. Write fractions in simplest form and show your work.

1. Write as a decimal: $\frac{5}{8}$

= 0.625

2. Write as a fraction: 0.75

= $\frac{3}{4}$

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Classify the number as Rational (R) or Irrational (IR).

3. Classify: $\sqrt{169}$

Classification: R

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Convert each number as directed. Write fractions in simplest form and show your work.

4. Write as a fraction: $0.\overline{3}$ (the 3 repeats)

= $\frac{1}{3}$

5. Write as a decimal: $2\frac{1}{3}$ (2 and $\frac{1}{3}$)

= 2.333...

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{50}$

Classification: IR

Now simplify each expression first, then classify the result.

7. Classify: $\sqrt{2} \times \sqrt{8}$

Classification: R

8. Classify: $\sqrt{75} \div \sqrt{3}$

Classification: R

1.3 Estimating Non-Perfect Square Roots

NC.8.NS.2 Use rational approximations of irrational numbers to compare the size of irrational numbers and locate them approximately on a number line.

Estimate each value to the **nearest tenth**, and plot it on the number line.

9. Estimate: $\sqrt{28}$



≈ 5.3

10. Estimate: $\sqrt{60}$



≈ 7.7

SCORE 5.0 Advanced Content

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Evaluate or simplify each expression if possible, *then* classify the result as Rational (R) or Irrational (IR).

11. Classify: $2 + \sqrt{16}$

Classification: R

12. Classify: $5 + \sqrt{7}$

Classification: IR

1.4 Ordering Real Numbers

NC.8.NS.2 Use rational approximations of irrational numbers to compare the size of irrational numbers and locate them approximately on a number line.

Order the set from **least to greatest**.

13. Set: $\{ 0.8, \sqrt{3}, 1.2, 7/5 \}$



Order = 0.8, 1.2, 7/5, $\sqrt{3}$

1.5 Area

A square garden has an area of 196 ft². Find the side length and the perimeter. Show all work.

$$A = 196 \text{ ft}^2$$

14.

Side length = 14 ft

Perimeter = 56 ft